

Title

Project Report

MSc Research Project / (Research) Practicum / Internship
Programme Name

Forename Surname

Student ID: XXX

School of Computing
National College of Ireland

Supervisor: XXX

September 2026

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XXX

*Abstract This project introduces a novel approach to 'Serverless Function Cold Start Mitigation using Predictive Auto-scaling'. We address the gap in Wang et al. (2025) where While keeping functions warm statically works, it wastes cost. This research implements predictive provisioning based on temporal access patterns.. Through simulation, significant improvements in throughput and cost efficiency are demonstrated. Introduction Cloud computing represents a shift. We investigate 'Serverless Function Cold Start Mitigation using Predictive Auto-scaling'.

In reference to Lee (2026), the integration highlights a distinct improvement metric.

Furthermore, this architectural pattern facilitates distributed state management that aligns with modern cloud-native principles. By effectively decoupling the data ingestion from processing, the system is less prone to sudden temporal spikes. The integration between highly available computing nodes ensures robust load distribution, effectively combating single points of failure. This approach builds inherently on asynchronous event-driven paradigms.

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[IMAGE PLACEHOLDER: terminal_output.png] *Conclusion* In conclusion, we successfully validated *scaling'*.

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